5G-based IoT Gateway with Edge ML

Course: Distributed Systems (521290S)
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Group # 20	
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Motivation and Overview

Problem

- Rapid growth of IoT devices generates massive volumes of sensor data.
- Traditional cloud-only architectures introduce latency and bandwidth overhead.
- Real-time applications require low-latency edge processing.

Solution

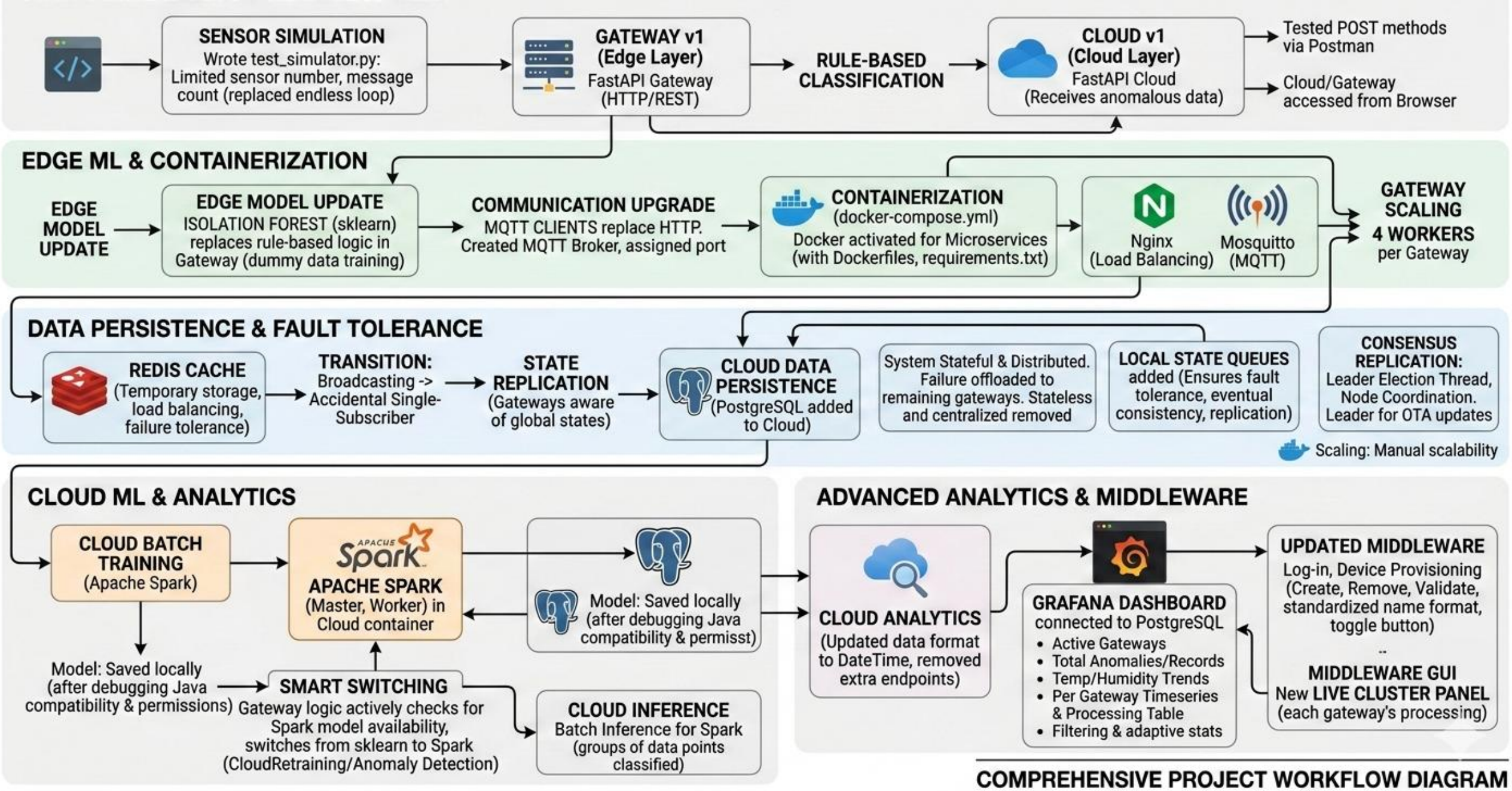
- Sensor processing big data and detecting anomalies using **edge and batch ML**.
- Using light-weight communication for **scalable and fault-tolerant** data streaming.
- Storing historical data in persistent **cloud storage**
- Visualising system performance and data analytics using **dashboarding.**

Deliverables and Implementation

The project implements a **distributed, microservices-based IoT** platform spanning edge devices and cloud analytics.

Key deliverables include:

- **IoT Device Simulator** generating thousands of sensor data.
- **MQTT protocol** between devices and gateways for scalable and consistent communication.
- **Distributed Edge Gateways** performing real-time anomaly detection and consensus.
- **Cloud Storage using PostgreSQL** for persistent data storage.
- **Apache Spark Cluster** for batch machine learning training.
- **Middleware Layer** for device management and gateway monitoring.
- **Grafana Dashboard** for real-time cloud analytics and visualisation.
- **Docker-based microservices architecture** for scalable deployment.



Distributed Architecture Overview

Docker Containers, Device Simulation and Gateway Logs

The system is deployed using **Docker containers**, enabling modular microservices for the following system components:

- **Device Simulator** generating sensor data (temp. and humidity).
- **MQTT Broker** handling publish–subscribe communication.
- **Load balancer** distributing workload across scalable gateways.
- Gateway logic for **leader election** and **state replication**.
- Edge **anomaly detection** by adaptive model selection.
- **Throughput, Delivery Rate** and **Latency** calculation.

```
(DS2026) PS D:\University-Masters\Semester_2\P3_Distributed_Systems\Lab\
gateway-2  | [cf445fc8ebe2] State recovered.
gateway-2  | Loading fallback sklearn model...
gateway-2  | ⚠️Fallback sklearn model loaded.
gateway-2  | Worker 0 started
gateway-2  | Worker 1 started
gateway-2  | Worker 2 started
gateway-2  | Worker 3 started
gateway-2  | State sync thread started
gateway-2  | Leader election thread started
gateway-2  | Spark batch processor thread started
gateway-2  | Model update checker thread started
gateway-2  | MQTT subscriber started for cf445fc8ebe2
gateway-2  | [cf445fc8ebe2] Leader is 48ca66107c8e
gateway-2  | [cf445fc8ebe2] Connected to MQTT with rc=0
gateway-2  | [cf445fc8ebe2] Leader is 48ca66107c8e
```

```
20 → published
86 → published
59 → published
86 → published
59 → published
86 → published
86 → published

✅ Simulation finished.
Started at : 2026-03-05T11:07:18.240906
Finished at: 2026-03-05T11:18:32.192588
Total duration: 673.95 seconds
Total messages sent: 50000
Throughput: 74.19 messages/second
```

project	-	-	-		2.43%	989.37MB /
postgres-1	cee803cf1cb6	postgres:1	5433:5432		0.04%	82.75MB /
redis-1	303717e74cc9	redis:7	6379:6379		0.32%	7.23MB / 7
mqtt-broker-1	f60740a43a35	eclipse-mo	1883:1883		0.04%	3.8MB / 7.6
spark-master	2cf9b0393408	apache/sp	7077:7077		0.13%	253.5MB /
gateway-1	48ca66107c8e	project-gat			0.46%	152.4MB /
grafana	415df54e540a	grafana/gr	3000:3000		0.69%	96.05MB /
cloud-1	fa66be8db815	project-clo	9000:9000		0.45%	105.4MB /
spark-worker	b48b94ae438d	apache/sp			0.11%	214.8MB /
nginx-1	1a4911e8319a	nginx:lates	8080:80		0%	3.44MB / 7
middleware-1	0f21035a0457	project-mic	7000:7000		0.19%	70MB / 7.6

Middleware and PostgreSQL Database

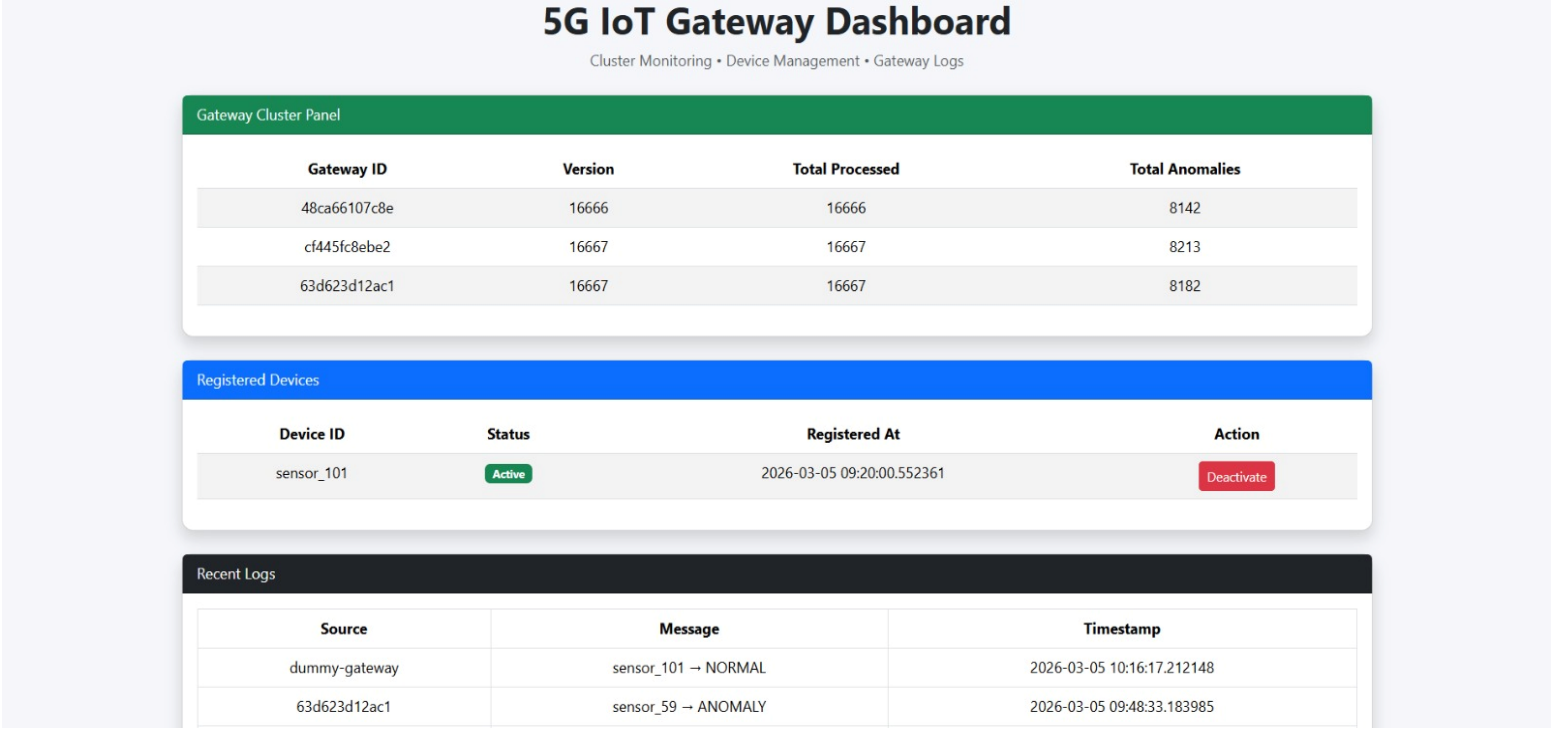
All sensor data is eventually stored in a **PostgreSQL cloud database**.

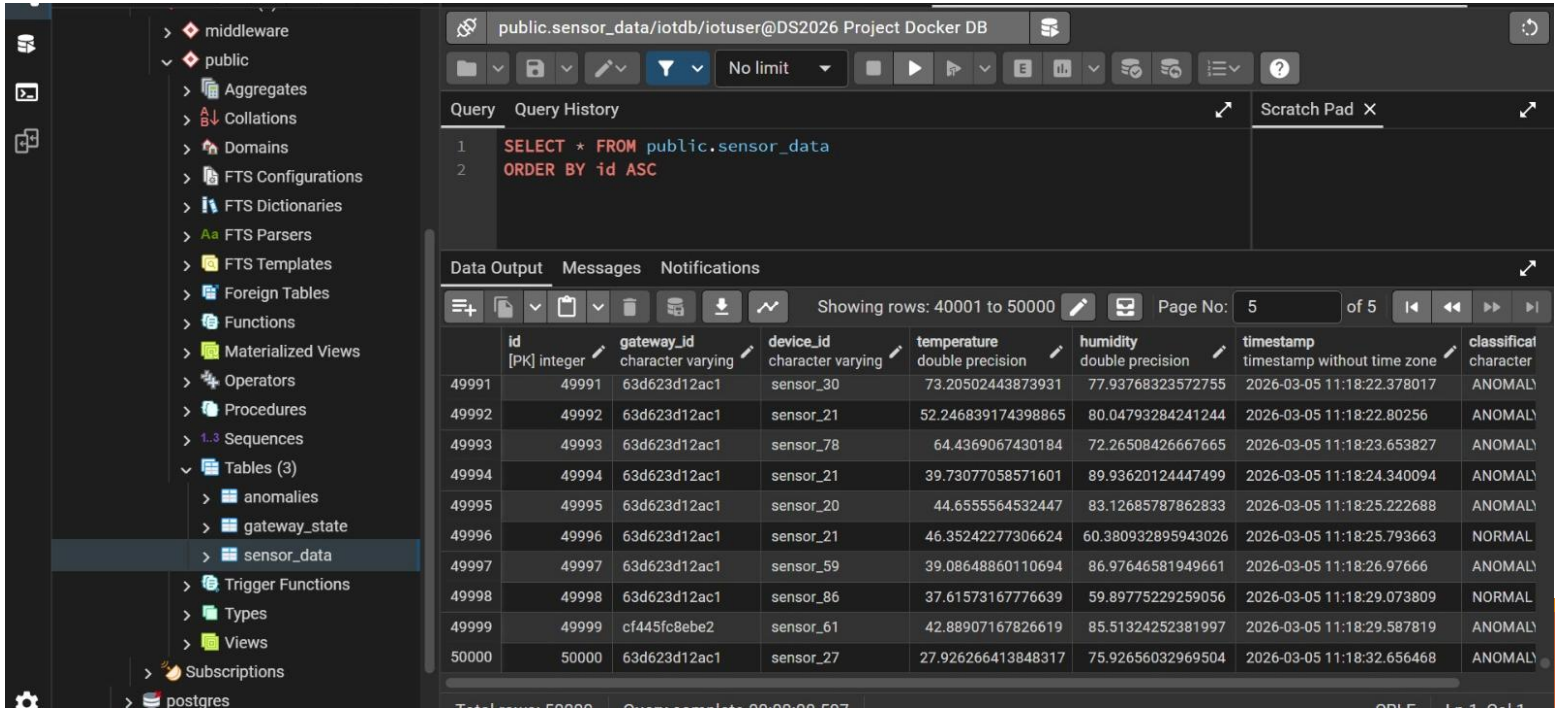
Database tables store:

- Sensor readings
- Gateway states
- Anomaly detection results
- Gateway processing logs

A **middleware layer** provides administrative functionality:

- Device provisioning
- Authentication and access control
- System logging
- Gateway cluster monitoring





Edge and Cloud ML with Spark

At the **Edge (Gateway):**

- Lightweight **Isolation Forest model**
- Performs **real-time anomaly detection**
- Enables **low-latency predictions**

```
gateway-4  | [d64fa68e10cd] Worker 3 → NORMAL (sklearn)
gateway-4  | [d64fa68e10cd] Worker 0 → ANOMALY (sklearn)
gateway-4  | [d64fa68e10cd] Leader is 0c24be04c212
gateway-4  | [d64fa68e10cd] Worker 3 → NORMAL (sklearn)
gateway-4  | [d64fa68e10cd] Worker 0 → NORMAL (sklearn)
gateway-1  | Spark batch processor thread started
gateway-1  | Model update checker thread started
gateway-2  | [0c24be04c212] Worker 3 → NORMAL (sklearn)
gateway-2  | [0c24be04c212] Worker 2 → ANOMALY (sklearn)
```

```
gateway-2  | MQTT subscriber started for 0507cc788898
gateway-2  | [0507cc788898] Leader is 056bda2cb568
gateway-2  | [0507cc788898] I am LEADER
gateway-2  | 🔄 New Spark model detected.
gateway-2  | Loading Spark model...
gateway-2  | [0507cc788898] I am LEADER
gateway-2  | [0507cc788898] I am LEADER
gateway-2  | ✅ Spark model loaded successfully.
gateway-2  | Model upgraded to Spark version.
gateway-2  | [0507cc788898] Worker 3 → ANOMALY (spark)
gateway-2  | [0507cc788898] Worker 3 → NORMAL (spark)
gateway-2  | [0507cc788898] Worker 3 → NORMAL (spark)
```

At the **Cloud:**

- Trained on historical sensor data.
- **Apache Spark cluster** performs distributed batch training using **Random Forest**
- Trained models are **redeployed to edge gateways**

Spark 3.5.1 **Spark Master at spark://172.19.0.5:7077**

URL: spark://172.19.0.5:7077
Alive Workers: 1
Cores in use: 8 Total, 0 Used
Memory in use: 6.6 GiB Total, 0.0 B Used
Resources in use:
Applications: 0 Running, 1 Completed
Drivers: 0 Running, 0 Completed
Status: ALIVE

Workers (1)

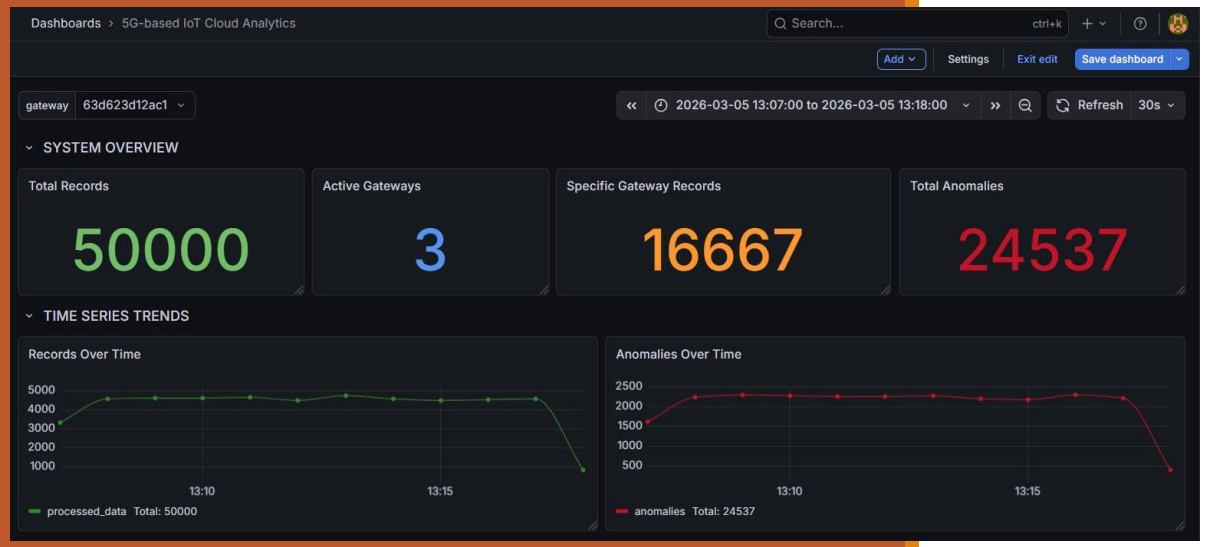
Worker Id	Address	State	Cores	Memory	Resources
worker-20260305152654-172.19.0.8-37015	172.19.0.8:37015	ALIVE	8 (0 Used)	6.6 GiB (0.0 B Used)	

Running Applications (0)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration

Completed Applications (1)

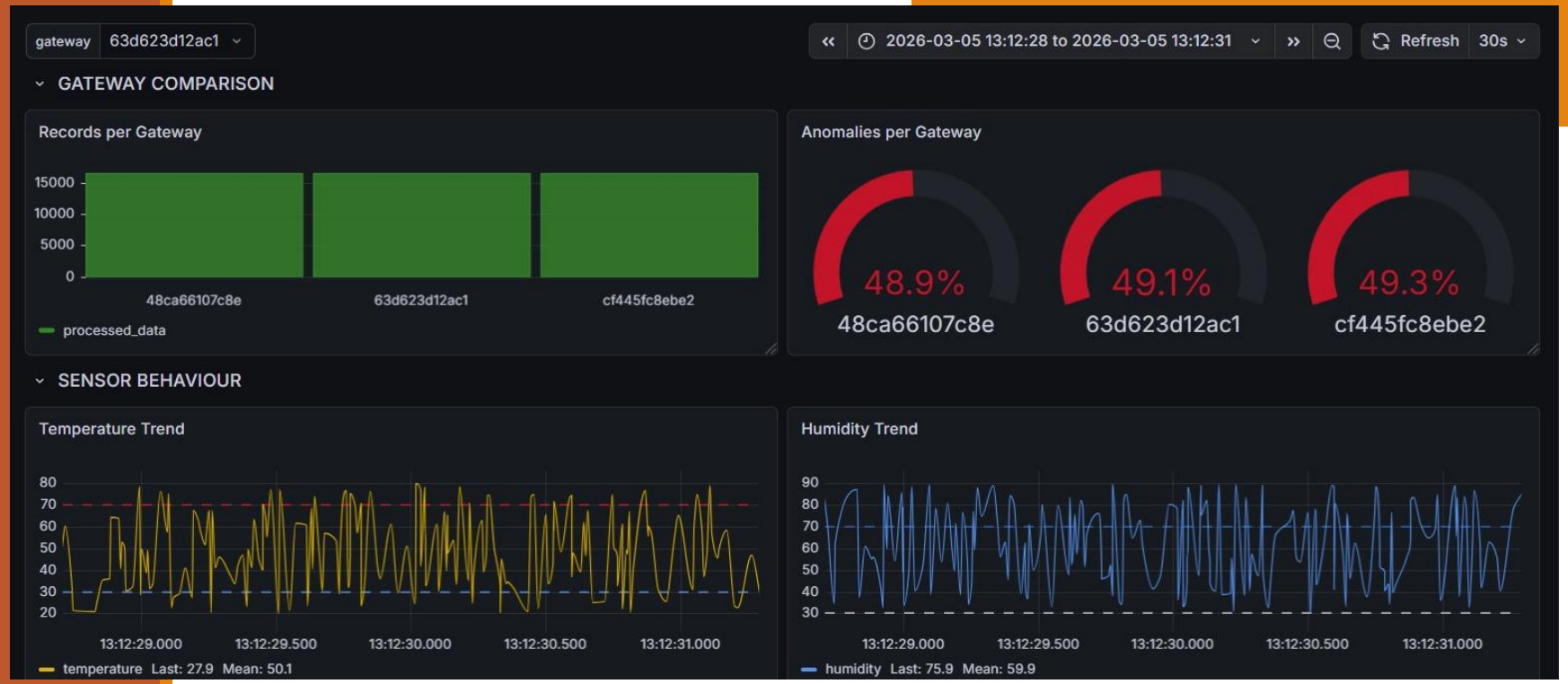
Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20260305153210-0000	IoTBatchTraining	8	1024.0 MiB		2026/03/05 15:32:10	root	FINISHED	1.3 min



Cloud Analytics with Grafana

Dashboards connected to PostgreSQL provide visual insights:

- Sensor data trends
- Detected anomalies
- Per Gateway trends
- System activity metrics



Metric	Result	Status
Latency (Low Payload + Messages)	~0.04 sec	Very Low
Latency (High Payload + Messages)	~0.47 sec	Low even with 50,000 messages
Delivery Rate (Cloud Failure)	100%	Perfect
Delivery Rate (Gateway Failure)	~89%	Fault tolerant
Throughput (1 Gateway)	73 msg/sec	High
Throughput (5 Gateways)	~69 msg/sec	Scalable

System Performance and Evaluation

Learning Outcomes:

- Distributed systems architecture
- Docker-based microservices deployment
- Edge machine learning inference
- Distributed inference with Apache Spark
- Real-time analytics using Grafana

Future Work:

- Automatic gateway scaling
- Improved distributed consensus and fault tolerance
- Adaptive ML model selection based on the payload

GITHUB REPOSITORY

HTTPS://GITHUB.COM/FAIZANTANWIR/DS2026-PROJECT-GROUP-20.GIT